

C4: Yingchuan parameter checkpoint $\lambda=0.4$, $\beta=0.001$

Sapelo array 48481042; five prespecified exploratory seeds. This is one of eight sequential checkpoints. All five jobs are terminal before this report; no next beta was launched for its construction.

Evidence boundary: measured CIFAR-100-C severity-5 trajectories and finite intervention responses. Numerical failure is a failed computation, not behavioral deterioration. A finite F response does not identify a closed Jacobian, spectrum or phase transition.

Complete five-seed status

Seed	Parent blocks	Extension blocks	Forks verified	H2O analyses	Coverage / reason
202	15	30	4/4	2/2	complete
303	15	30	4/4	2/2	complete
404	15	30	4/4	2/2	complete
505	15	30	4/4	2/2	complete
606	15	30	4/4	2/2	complete

A component is reported only after its own completion and verification. Parent, extension, fork and H2O gaps remain explicit; the raw run and failed attempts remain on Sapelo.

Exploratory readout: visit-3 minus visit-1 same-domain CE is positive in 4/5 seeds (mean +0.090); classification error changes by -3.94 percentage points on average. At H1410, 5/10 correlated parent nodes have harmful closed acceptance and 7/10 have positive feedback interaction. The signs are mixed; these are descriptive, not a persistent-deterioration or stability claim.

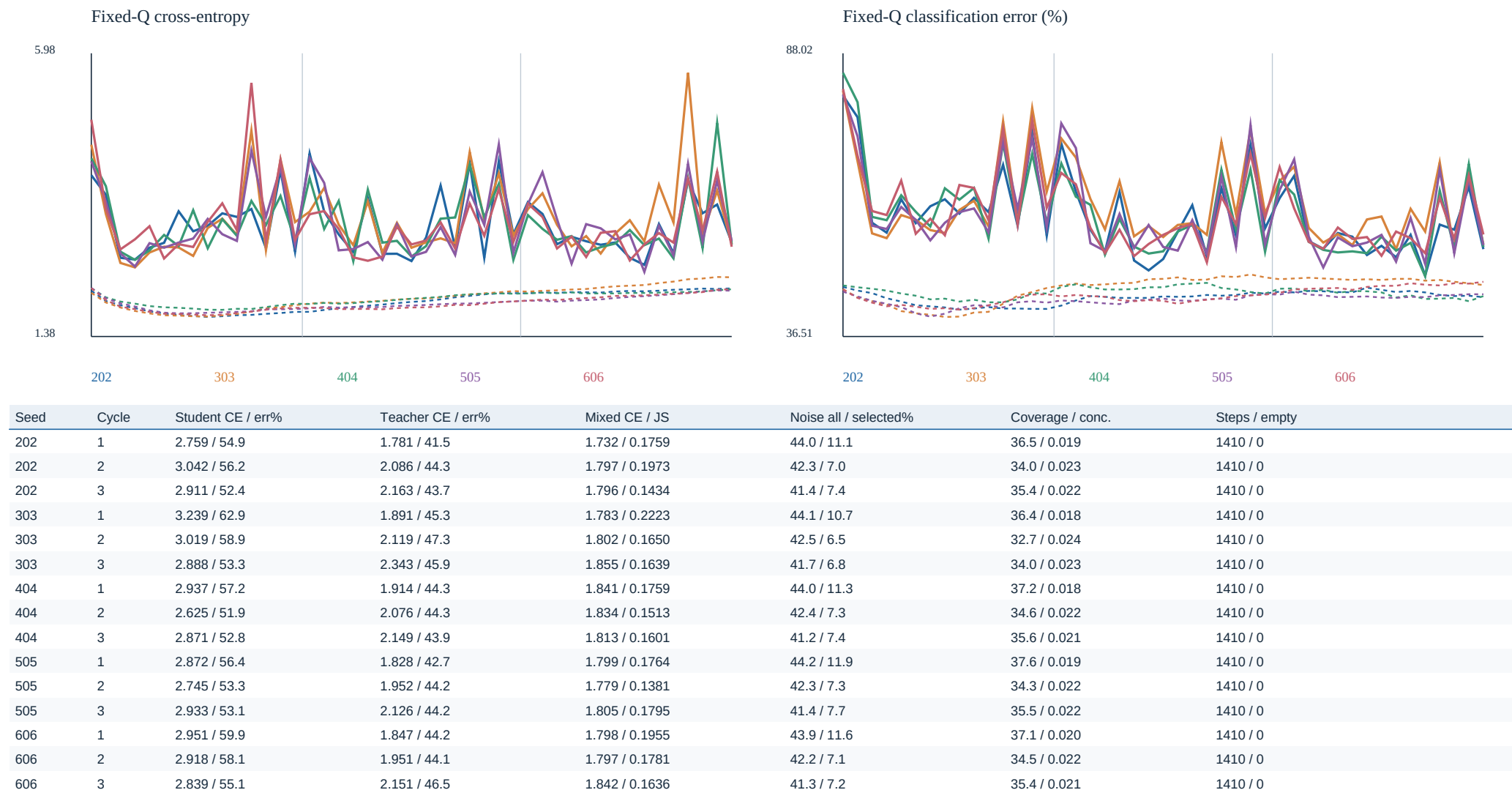
C4-1. Proxy versus labelled local update

Negative post-minus-pre CE means local improvement. Near zero means either absolute change is below $1e-4$. Natural C4 executes updates without a separate accept/reject evaluator.

Seed	Probes	Both improve	Proxy down / local up	Proxy up / local down	Both worsen	Near zero
202	90	43	46	0	0	1
303	90	51	39	0	0	0
404	90	58	30	0	2	0
505	90	61	29	0	0	0
606	90	56	33	0	1	0

C4-2. Complete fixed-Q trajectories and pseudo-supervision

The fixed-Q panel uses the same 1,500 image-corruption evaluations at every block. Curves are unsmoothed. Dotted curves are teachers; solid curves are students. Missing/failed cells are absent, not assigned zero risk.



Cycle CSV retains teacher/student disagreement, mixed-target confidence, count-weighted pseudo-label statistics, selected-class concentration and elapsed block time. Selected noise is undefined when no examples were selected.

C4-3. Same-domain returns

Visit-3 minus visit-1 post-block student CE for each named corruption. Positive means worse on that domain; all domains and all terminal cells are retained. The companion block CSV includes pre/post CE and error for student and teacher at every visit.

Corruption	202	303	404	505	606
gaussian_noise	0.231	-0.972	-1.001	-0.021	-1.759
shot_noise	-0.394	0.483	-0.463	0.234	-0.452
impulse_noise	0.576	0.923	0.538	0.703	0.314
defocus_blur	0.491	0.202	0.271	0.336	0.490
glass_blur	-0.034	0.114	-0.015	0.609	-0.234
motion_blur	0.432	0.377	0.262	0.222	0.484
zoom_blur	0.199	0.343	0.325	0.120	0.219
snow	-0.650	0.758	0.175	0.553	-0.190
frost	-0.717	-1.128	0.187	-0.209	-0.080
fog	0.005	0.304	0.259	0.299	0.039
brightness	0.152	0.349	0.167	0.066	0.127
contrast	0.264	0.106	0.033	0.496	-0.450
elastic_transform	0.704	0.255	0.052	0.017	0.432
pixelate	-0.462	-0.392	1.304	0.381	-1.117
jpeg_compression	0.073	-0.416	0.215	0.403	0.249

C4-4. Per-seed parameter summary

Seed	Mean post CE / err%	Mean return dCE / dErr pp	Cycle-3 CE / err%	Coverage%	Updates	Block cost min
202	2.692 / 52.4	0.058 / -5.51	2.691 / 49.9	35.3	4230	15.3
303	2.767 / 53.7	0.087 / -3.11	2.817 / 52.1	34.4	4230	14.9
404	2.677 / 52.2	0.154 / -4.00	2.718 / 50.3	35.8	4230	15.1
505	2.757 / 53.0	0.281 / -1.75	2.880 / 52.0	35.8	4230	15.3
606	2.672 / 52.2	-0.129 / -5.33	2.623 / 49.7	35.7	4230	15.4

C4-5. Matched accept/reject and closed/replay forks

The first nonempty update after preselected blocks 5 and 15 is the focal candidate. Positive accept-minus-reject CE is worse. $A_{\text{closed}} = \text{accept_closed} - \text{reject_closed}$; $A_{\text{open}} = \text{accept_replay} - \text{reject_replay}$; $I_{\text{feedback}} = A_{\text{closed}} - A_{\text{open}}$. The H94 and H1410 forks share their initial state and schedule; their latest common scored checkpoint, H93, agrees exactly. H94 is therefore a correlated branch, not an independent replicate. The H1410 run does not separately score H94.

Seed	Parent	Immediate Q dCE / dErr pp	Immediate domain dCE / dErr pp
202	5	0.043 / 0.73	-0.001 / -0.33
202	15	-0.005 / -0.47	0.120 / 1.33
303	5	0.020 / 0.73	-0.008 / -0.67
303	15	-0.044 / -1.07	-0.262 / -2.27
404	5	-0.072 / -1.53	-0.124 / -1.40
404	15	0.164 / 0.67	0.538 / 2.93
505	5	0.020 / -0.87	-0.050 / -1.47
505	15	0.022 / 0.73	0.151 / 1.73
606	5	-0.019 / -0.27	0.003 / 0.33
606	15	-0.052 / -0.80	-0.058 / -1.27

H94 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_{closed}	A_{open}	I_{feedback}
202	5	2.802	2.802	2.844	2.740	0.041	-0.062	0.103
202	15	4.228	4.228	4.377	3.840	0.149	-0.388	0.537
303	5	2.888	2.888	2.888	2.923	0.001	0.035	-0.035
303	15	3.820	3.820	3.352	3.766	-0.468	-0.054	-0.414
404	5	2.950	2.950	2.911	2.934	-0.039	-0.016	-0.023
404	15	3.429	3.429	4.180	4.336	0.752	0.908	-0.156
505	5	2.694	2.694	2.853	2.866	0.159	0.173	-0.014
505	15	3.545	3.545	4.485	4.345	0.940	0.801	0.139
606	5	2.806	2.806	2.673	2.880	-0.133	0.074	-0.207
606	15	3.150	3.150	3.342	3.169	0.192	0.019	0.173

H1409 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.263	3.263	3.626	2.737	0.364	-0.526	0.890
202	15	2.715	2.715	3.042	2.962	0.326	0.246	0.080
303	5	3.582	3.582	3.574	3.157	-0.007	-0.425	0.418
303	15	2.790	2.790	3.019	2.873	0.230	0.083	0.147
404	5	3.002	3.002	3.766	3.003	0.764	0.001	0.763
404	15	2.781	2.781	2.625	2.748	-0.156	-0.034	-0.122
505	5	3.012	3.012	2.915	2.807	-0.098	-0.206	0.108
505	15	3.041	3.041	2.745	2.678	-0.296	-0.363	0.067
606	5	2.823	2.823	2.611	2.908	-0.212	0.086	-0.297
606	15	3.222	3.222	2.918	2.763	-0.304	-0.459	0.154

H1410 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.080	3.080	3.415	2.714	0.335	-0.366	0.701
202	15	2.799	2.799	3.073	3.051	0.274	0.252	0.022
303	5	3.728	3.728	3.909	3.235	0.182	-0.492	0.674
303	15	2.845	2.845	3.075	2.880	0.231	0.035	0.195
404	5	3.011	3.011	3.827	2.977	0.816	-0.034	0.849
404	15	2.747	2.747	2.624	2.790	-0.123	0.043	-0.166
505	5	2.952	2.952	2.868	2.902	-0.084	-0.050	-0.034
505	15	3.069	3.069	2.837	2.687	-0.232	-0.382	0.150
606	5	2.757	2.757	2.565	2.901	-0.192	0.144	-0.337
606	15	3.251	3.251	2.998	2.773	-0.253	-0.478	0.225

C4-6. Diagnostic state, finite F and local identifiability

x=(student fixed-Q CE, mixed-target fixed-Q CE, student/teacher JS, mixed confidence). F is the mean H20 future x over three base tapes at each observed parent; it is a conditional response, not a closed transition law. The block CSV stores every observed x_t .

Seed	Parent	x0	F_H20(x0)	Tape SD	Finite	J status
202	5	2.823 / 1.858 / 0.243 / 0.674	2.796 / 1.846 / 0.231 / 0.673	0.052 / 0.001 / 0.007 / 0.000	132/132	not identified
202	15	2.759 / 1.732 / 0.176 / 0.653	3.328 / 1.731 / 0.279 / 0.653	0.103 / 0.001 / 0.025 / 0.000	132/132	not identified
303	5	2.745 / 1.864 / 0.228 / 0.679	2.513 / 1.853 / 0.198 / 0.678	0.088 / 0.001 / 0.005 / 0.000	132/132	not identified
303	15	3.239 / 1.783 / 0.222 / 0.644	3.206 / 1.783 / 0.259 / 0.644	0.096 / 0.001 / 0.026 / 0.000	132/132	not identified
404	5	2.778 / 1.933 / 0.242 / 0.677	2.768 / 1.925 / 0.218 / 0.676	0.204 / 0.001 / 0.010 / 0.000	132/132	not identified
404	15	2.937 / 1.841 / 0.176 / 0.657	3.449 / 1.840 / 0.279 / 0.657	0.213 / 0.001 / 0.019 / 0.000	132/132	not identified
505	5	2.896 / 1.908 / 0.250 / 0.685	2.692 / 1.896 / 0.222 / 0.684	0.173 / 0.001 / 0.020 / 0.000	132/132	not identified
505	15	2.872 / 1.799 / 0.176 / 0.656	3.373 / 1.800 / 0.260 / 0.656	0.117 / 0.000 / 0.013 / 0.000	132/132	not identified
606	5	3.172 / 1.876 / 0.269 / 0.680	2.824 / 1.867 / 0.219 / 0.679	0.181 / 0.000 / 0.018 / 0.000	132/132	not identified
606	15	2.951 / 1.798 / 0.195 / 0.653	3.534 / 1.796 / 0.254 / 0.653	0.473 / 0.001 / 0.020 / 0.000	132/132	not identified

Perturbation, conditioning and held-out diagnostics

Seed	Parent	Median G	Rank	Condition	Scale diff	Held-out median	Closure	Reason
202	5	26.2	4	9.3	1.50	71.92	falsified	not established
202	15	37.3	4	5.7	0.82	293.05	falsified	not established
303	5	19.7	4	2.5	1.34	70.60	falsified	not established
303	15	30.5	4	4.6	1.21	93.83	falsified	not established
404	5	21.4	4	4.0	1.18	48.68	falsified	not established
404	15	47.5	4	3.3	1.03	226.25	falsified	not established
505	5	44.9	4	2.4	0.95	153.77	falsified	not established
505	15	38.6	4	2.4	0.59	249.11	falsified	not established
606	5	12.3	4	3.4	0.97	53.20	falsified	not established
606	15	36.8	4	14.0	1.01	175.85	falsified	not established

Independent student/teacher perturbations, input rank/condition, two scales, held-out directions and momentum closure checks are retained in the compact analysis files. No eigenvalue or phase boundary is claimed without a closed and predictive state law.

C4-7. Bounded parameter-map reading

This checkpoint isolates one pair $\lambda=0.4$, $\beta=0.001$ with five exploratory seeds. Seeds are reused across later pairs for paired comparison; they are not independent cross-parameter replications. Outcomes at this pair alone cannot locate a stability boundary. Numerical failures and improvement/null findings remain in the reported denominator.

Observed coverage: 5/5 complete 45-block trajectories, 10 verified H1410 parent nodes, 5 positive A_{closed} and 7 positive I_{feedback} . Parent nodes within one seed are correlated. A positive endpoint or interaction alone does not establish sustained recursive deterioration.

Coverage ledger and missingness

C4 item	Present evidence	Missing / limitation
C4-1	450 completed local probes	Unavailable after failed/incomplete trajectories
C4-2	5 full trajectories and cycle process records	Failed/partial cells not extrapolated
C4-3	75 same-domain return comparisons	Only complete first/third visits
C4-4	5 per-seed parameter summaries	One pair cannot characterize a sweep
C4-5	30 fork horizons	Unavailable parent nodes explicitly excluded
C4-6	10 H20 parent analyses	J/spectrum only if closure and held-out criteria pass
C4-7	One-pair status/sign description	No identified region or boundary

Provenance: Sapelo 48481042, compact evidence MANIFEST.json and AUDIT.json. Raw checkpoints, logits, step tapes and all failures remain under /scratch/yz54720/RSI2RSD/C4/sweep_yingchuan_20260923/results/I04_b0001/. The reserved confirmation panel was not read.